FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

COMPUTER SCIENCE

Past Paper 2024 • Paper Code: 2HA-I 24007 — 40072 (Version B) / 42072 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

COMPUTER SCIENCE HSSC-II — SECTION – A (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 13 | **Paper Code:** 40072 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. The technique that reuses an existing class to build a new class is known as:

(A) Polymorphism
(B) Inheritance
(C) Data hiding
(D) Function overloading

Q2. The program in execution is called:

(A) Web browser
(B) File
(C) Kernel
(D) Process

Q3. In flow chart the (parallelogram) symbol is used to represent: (A parallelogram symbol is printed in the source scan.)

(A) Arithmetic operations
(B) Start
(C) Decision
(D) Input/output operations

Q4. The step by step procedure to solve a problem is called:

(A) Direct implementation
(B) Algorithm
(C) Flow chart
(D) Coding

Q5. In file handling stream is a:

(A) Series of variable address
(B) Series of variables
(C) Series of bytes
(D) Series of constants

Q6. A type of member function that is preceded by tilde sign ~ is called:

(A) Polymorphism
(B) Destructor
(C) Object
(D) Constructor

Q7. What will be the output of the following expression? int x=20, y=3; cout<<x%y;

(A) 2
(B) 6
(C) 0
(D) 5

Q8. Which of the following loops is called post-test loop?

(A) Nested for loop
(B) For loop
(C) While loop
(D) Do-while loop

Q9. Which of the following is correct syntax of string declaration?

(A) char book[30];
(B) int book[30];
(C) char book[30];
(D) char book{ };

Q10. int arr[5]={9,10,12,19,98}; What will be stored at arr[4];

(A) 10
(B) 9
(C) 98
(D) 19

Q11. The parameters used in function call are called:

(A) Local parameters
(B) Formal parameters
(C) Global parameters
(D) Actual parameters

Q12. Which of the following functions is used to find if the control has reached the end of file or not?

(A) eof( )
(B) bof( )
(C) gets( )
(D) puts( )

Q13. The ability to use an operator or function in multiple ways is called:

(A) Virtual function
(B) Inheritance
(C) Polymorphism
(D) Data hiding

COMPUTER SCIENCE HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:40 Hours | **Maximum Marks:** 62 | **Paper Code:** 2HA-I 24007(B)

Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [03] Write down one application for each of the following types of operating systems. a. Time-sharing b. Real time c. Embedded — OR (03) — Write down any three differences between process and thread.

(ii) [03] How is protection system important in an operating system? — OR (03) — Write down any three responsibilities of system analyst.

(iii) [03] Briefly describe any three types of deployment phase. — OR (03) — Briefly describe the five states of process with diagram.

(iv) [03] Write down any three differences between if-else-if and switch statements. — OR (03) — What is understood by functional and non-functional requirements in requirement gathering phase of SDLC?

(v) [03] What will be the output of following code segment? int i=3; int j=i++; int k=++i; cout<<i<<" "<<j<<" "<<k; — OR (03) — What will be the output of the following code segment? for(int i=0; i<3;i++) { cout<< "Enter a character"; cin>>ch; out.put(ch); }

(vi) [03] What will be the output of the following code? int m=5, c=0; while(c<100) { m=m+1; c=c+1; } cout<<m; — OR (03) — Write down the output of the following code segment. {int a=7, b=12; fun(a,b); cout<<a<<" "<<b; } void fun(int m,int &n) { n=n-2*m; m=2*m; }

(vii) [03] int arr[3][4]={{12,0,5,10},{7,8,19,30},{33,1,2,22}}; Write down a statement that replaces 19 with 50. — OR (03) — Write a program that inputs a string and displays the number of character in it.

(viii) [03] Fill in the following table with Yes/No. Access / Public / Private: Access of members in same class; Access of members in derived class; Access of members outside the class — OR (03) — Consider the following array declaration: int arr[8]; a. How many elements are there in the array? b. Determine the highest index of the array. c. Determine the lowest index of the array.

(ix) [1+2] What are strings? Briefly explain how to declare a string. — OR (03) — Differentiate between local and static variables.

(x) [03] Compare binary files with text files in light of file handling in C++. — OR (03) — Briefly describe any three modes of file opening in C++.

(xi) [03] What is the use of reference operator (&) in pointers? — OR (03) — How is an array different from a simple variable? Explain with an example.

(xii) [03] Write down the syntax of using the EOF function in file handling. — OR (03) — Briefly explain the concept of data hiding in classes using an example.

(xiii) [03] Write down any two advantages and one disadvantage of function overloading. — OR (03) — Briefly explain the following terms related to array: a. Size of array b. Name of array c. Index.

(xiv) [03] Explain the role of Project manager in SDLC. — OR (03) — What is the use of break statement in C++?

> Note: SECTION – C (Marks 20) — Note: Attempt the following questions. (5x4=20)

Q3. (Marks 05) Q.3 Write down a C++ program that finds the area and perimeter of a square. — OR (3+1+1) — Why we use pointers? What is the difference between pointer variable declaration and initialization? Explain with example.

Q4. (Marks 05) Q.4 Write a C++ program that contains two integer data members which are initialized to 100 when an object is created. It has a member function avg () that displays the average of data members. — OR (3+2) — Explain the difference between constant and variable with the help of programming examples.

Q5. (Marks 05) Q.5 Explain the use of following in C++ by using examples. • Continue statement • Exit function — OR (2+2+1) — Write down the importance of using functions in C++. Also explain the following components of function: • Function definition • Function call.

Q6. (Marks 05) Q.6 Write a C++ program that inputs a positive integer and prints whether it is prime or composite. — OR (1+4) — Explain the concept of function signature. Also write down any four advantages of using functions in C++.

VERSION D

COMPUTER SCIENCE HSSC-II — SECTION – A (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 13 | **Paper Code:** 42072 (Version D)

> Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

> حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

> Note: Fill the relevant bubble against each question according to curriculum:

> ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. An object is a variable of type:

(A) class
(B) int
(C) float
(D) file

Q2. Which of the following is a system software needed to load applications/programs into the memory and execute it?

(A) Language processor
(B) Device driver
(C) Operating system
(D) Utility program

Q3. Monitoring the position of a rocket in the space is the application of _____ operating system.

(A) Real-Time
(B) Time-sharing
(C) Batch processing
(D) Multi-programming

Q4. Which of the following symbols in flowchart represents decision making? (Flowchart symbols are printed in the source scan.)

(A) rectangle symbol
(B) parallelogram symbol
(C) diamond symbol
(D) circle symbol

Q5. Which of the following operators is arithmetic assignment operator?

(A) ||
(B) +=
(C) +===
(D) !=

Q6. What will be the value of x after executing the following expression? int x,y=3; x=y++; cout<<x;

(A) 5
(B) 2
(C) 3
(D) 4

Q7. The statement in C++ that transfers the control to the start of the loop body is:

(A) Switch
(B) Continue
(C) Exit
(D) Break

Q8. What will be the output of the following expression? int a=15, b=10; a>b? a+b: a-b;

(A) 10
(B) 05
(C) 15
(D) 25

Q9. If int arr[2][3]={15,21,9,84,33,72}; what will be at arr[1][1];?

(A) 21
(B) 84
(C) 72
(D) 33

Q10. Which of the following functions is used to append the string 2 to the end of string 1?

(A) strlen( )
(B) strcat( )
(C) strcpy( )
(D) strcmp( )

Q11. The parameters used in the header of function definition are called:

(A) Global parameters
(B) Actual parameters
(C) Formal parameters
(D) Local parameters

Q12. The correct syntax to write single character in file is:

(A) obj.gets();
(B) obj.put(ch);
(C) obj.get(ch);
(D) obj.gets(ch);

Q13. Under which access specifier, the 'data members' and 'member functions' are accessible from outside the class?

(A) Private
(B) Public
(C) Protected
(D) Object

COMPUTER SCIENCE HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:40 Hours | **Maximum Marks:** 62 | **Paper Code:** 2HA-I 24007 (D)

> Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [03] Write down any three features of Multi-user Operating System. — OR (03) — How is an argument pass by value different from an argument pass by reference?

(ii) [03] Describe any three objectives of SDLC. — OR (03) — How is multi-threading different from multi-tasking? Give any three reasons.

(iii) [03] Briefly explain the three steps of requirement engineering phase of SDLC. — OR (03) — Write down any three differences between post-tested and pre-tested loops with example.

(iv) [03] Write down the use of string stream in file handling. — OR (03) — How is a project manager responsible for planning execution and closing of a project?

(v) [03] Rewrite the following for loop into do-while loop. for(int n=0;n<10;n++) cout<<n<<" "; — OR (2+1) — Briefly explain the concept of two-dimensional array with an example.

(vi) [03] Write down the purpose of sizeof( ) function in array. — OR (03) — Write down the output of the following code-segment. int p,q,r; p=10; q=3; if(p&q==3) r=0; else r=1; cout<<r;

(vii) [03] Write a C++ code that prints sum of following. int arr[10]={1,2,3,4,5,6,7,8,9,10}; — OR (2+1) — Differentiate between strlen( ) and strcat( ) functions with an example.

(viii) [03] Briefly explain how a pointer variable is declared by using an example. — OR (03) — Write down a piece of code in C++ that shows the use of inline function.

(ix) [03] Write a piece of code that shows how a string copies into another string using strcpy( ) function. — OR (03) — Given the array definition: float a[5]={1,2,3}; a. How many elements are there in the array? b. What are the values of the first and last elements? (1+2)

(x) [03] Differentiate local and global variables. — OR (03) — Briefly explain the concept of data hiding in C++.

(xi) [03] Explain the difference between the following statements if P is a pointer variable: cout<<P; cout<<*P; — OR (03) — Write down the use of function overloading in terms of: Number of arguments; Datatypes of arguments; Return types.

(xii) [03] How is constructor different from destructor? — OR (03) — Write a C++ code that reads 03 characters from user and stores them in a file.

(xiii) [03] How is a binary file different from a text file in C++? — OR (2+1) — Briefly explain the concept of polymorphism with daily life example.

(xiv) [03] Rewrite the following code by using the conditional operator: if(a==b) cout<< "equal"; else cout<< "Not equal"; — OR (03) — Write down the output of the following code: int i,j,k; for(i=0,j=2,k=1;i<=2;i++) cout<<i+j+k;

> Note: SECTION – C (Marks 20) — Note: Attempt the following questions. (5x4=20)

Q3. (Marks 1+4) Q.3 Why is feasibility study important? Discuss any four types of it. — OR (2+2+1) — Write down the use of type casting in C++. Briefly explain two types of casting with example.

Q4. (Marks 2+2+1) Q.4 Write down a C++ program by using class that input two values using a member function of a class named input( ), then display the sum of two values by using another member function named show( ). — OR (2+2+1) — What is the purpose of setw in C++? Also compare the use of setw and endl manipulators in C++.

Q5. (Marks 4+1) Q.5 Explain use of exit( ) function in C++ with an example. — OR (05) — Write a C++ program that prints all the positive odd numbers upto 30 skipping those that are divisible by 5 using continue statement.

Q6. (Marks 2+2+1) Q.6 Explain the concept of function. Also explain the use of function definition and function call with the help of examples. — OR (2+2+1) — Name four functions of operating system and explain any two functions in detail.